Patents

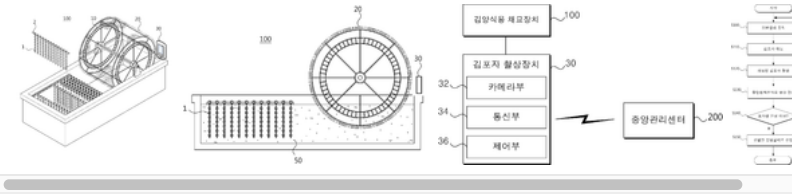
Methods for selecting spores from laver farm devices

Abstract

translated from Korean

In the present invention, an image pickup device mounted on one side of a harvesting device identifies and selects the amount of spores harvested in the Kimbal bridge, ultimately accumulating data on the state of the Kimbal valley and the amount of the spores being harvested, The present invention relates to a spore sorting system and method for selecting a spore sorting apparatus from a Kim, Yang-sik type mulching apparatus capable of automating a Kim, Yang-sik, which comprises a water tank provided with a Kimbal valley around which a net is wound, a frame provided on both sides of the lateral space of the water tank, A hanging rod which is installed at regular intervals between the frames, a hanging rod hanging at a predetermined interval to the hanging rod, and shells for implanting spores are bundled at regular intervals on the undergarment, An image pickup means for picking up spores collected in a Kimbal valley and an image pickup means for picking up Kim And a control center terminal for receiving the character image.
According to another aspect of the present invention, there is provided a washing machine comprising a water tank provided with a Kimbal valley around which a net is wound, a frame provided on both sides of the lateral space of the water tank, a latching body provided at a predetermined interval between the frames, Wherein the seedling harvesting apparatus comprises a chrysanthemum harvesting apparatus comprising a chrysanthemum harvesting apparatus and a harvesting apparatus for harvesting seedlings at regular intervals and shells for implanting spores, A method for screening spores from a spore sorting system from a spruce picking device, comprising the steps of: capturing a gimp from a shell to a gimbal bridge; photographing the captured spider by a gimp self-imaging device; And the amount of spores laid on the Kimgallgol from the transferred images, And a system.

Images (4)



Classifications

A01G33/00 Cultivation of seaweed or algae

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Life Sciences & Earth Sciences

Business, Economics & Management

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KR20190055413A

South Korea

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Other languages: Korean

Inventor: 조보현, 김동수, 조영후

Worldwide applications

2017 KR

Application KR1020170152108A events

2017-11-15	Application filed by 조보현, 김동수
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Info: Patent citations (3), Cited by (4), Legal events, Similar documents, Priority and Related Applications

External links: Espacenet, Global Dossier, Discuss

Claims (7)

Hide Dependent

translated from Korean

A frame provided on both sides of the frame in a lateral space portion of the water tub; a latching protrusion provided at a predetermined interval between the frames; and a hanging hook Rods and shells implanted with spores are bundled at regular intervals on the underline;
An image pickup means provided on one side of the steeping device for picking up spores collected in a Kimbal bridge; And
And a control center terminal for receiving the gimp image photographed by the image pickup means. The method according to claim 1,
Wherein the imaging means uses a water temperature lamp embedded in a biological microscope. The method according to claim 1,
The control center terminal,
A spore sorting system from Kim, Yong - Sik harvesting apparatus characterized by automating the verification of the completion of Kim Jong - The method of claim 3,
The big data includes:
A spore sorting system from the Kim Jung Sik harvesting apparatus, characterized by accumulating a large amount of image data in which the seedling planting is completed, and using the accumulated data to use a machine learning technique. A frame provided on both sides of the frame in a lateral space portion of the water tub; a latching protrusion

provided at a predetermined interval between the frames; and a hanging hook A spatula sorting device for picking up a spatula from Kim Jae-sik, which includes an apparatus for picking up a Kimbong sprout attached to one side of the apparatus, CLAIMS 1. A method for screening spores from a system,
A step of picking up gimp from the shells on the gimbal bridge;
A step of photographing the photographed gimpo self-imaging apparatus;
Transmitting the photographed gimp image to a control center terminal; And
Determining the amount of spores harvested in the Kimbig rope from the transmitted image, and sorting Kimbyeonggulle. 6. The method of claim 5,
Wherein the imaging device includes a mobile terminal equipped with a camera, and the camera includes a water temperature lamp embedded in a biological microscope. 6. The method of claim 5,
Wherein the image transmitted to the control center terminal is stored in a terminal database in a large amount of image data and the big data is formed using a machine learning technique using the stored data.

Description

translated from Korean

Technical Field [0001] The present invention relates to a spore sorting system and a method for selecting spores thereof from a strawberry harvesting apparatus,

More particularly, the present invention relates to a spore sorting system and method for picking up spoons from a Kimyeong-style harvesting apparatus, and more particularly, to an image picking apparatus mounted on one side of a harvesting apparatus, The present invention relates to a spore sorting system and method for sorting spores from a potato harvesting apparatus capable of automating the process by accumulating data on the state of Kimgulbyeol and the amount of sprouted spores,

Conventional manual method of planting seedlings has been limited to work efficiency because it is possible to confirm the completeness of planting materials by manually inputting all the work procedures and handing them by hand and checking them with eyes. , It is necessary to construct an automation system for Kim Jong-hyang plant to improve the accuracy and efficiency of planting work.

Also, At present, most of the domestic farms in Korea have planted seeds of Kim Jongmyo directly to the sea farm where Kim Bull was installed for the farming of Kimchi, which is good for economical efficiency such as planting efficiency and pollution of the sea and environmental protection. There is a disadvantage that it is not a method. It is necessary to develop an automation system for terminating the shore top to effectively improve it.

In order to solve the above problems, the object of the present invention is to develop a system for verifying the completeness of a nodule using Big Data and an apparatus for collecting an image of a nodule for mobile, System and management system.

The spore sorting system according to the present invention comprises a water tank provided with a Kimbal valley around which a net is wound, a frame provided on both sides of the lateral space of the water tank, A hanging rod body, a hanging rod hanging at a predetermined distance from the hanging rod body, and a shell embroidering a spore of the spore are bundled at regular intervals on the underarm row, a straw picking device installed on one side of the straw picking device, An imaging means for photographing spores, and a control center terminal for receiving the gimp image photographed by the imaging means.

Further, the imaging means can use a water temperature lamp embedded in a biological microscope.

In addition, the control center terminal can automate the verification of the completeness of the Kim Jongmyo by converting the images of Kimpo photographs into big data.

Also, the big data can accumulate a large amount of image data in which the seedling planting is completed, and utilize the machine learning technique utilizing the accumulated data.

The spore method comprises a water tank provided with a Kimbal valley around which a net is wound, a frame installed on both sides of the lateral space of the water tank, A hanging rod which hangs at a predetermined distance from the hanging rod, and shells for implanting spores are bundled at regular intervals on the underline; A method for sorting spores from a spore sorting system from a Kim Jae-sik harvesting apparatus comprising an imaging device for taking a picture, comprising the steps of: A step of transmitting a child image to the control center terminal, Determine the amount and character gimbal yoke includes a step of screening.

Further, the imaging device includes a mobile terminal equipped with a camera, and the camera may include a water temperature lamp embedded in a biological microscope.

Also, the image transmitted to the control center terminal can store a large amount of image data in the terminal database, and can be converted into big data using the machine learning technique using the accumulated data.

The present invention has the advantage of improving the efficiency and scalability of the nursery by automatically collecting the nursery planting state image of the nursery using the mobile imaging image capturing device to determine the nursery planting completeness.

Brief Description of Drawings Fig. 1 is an explanatory diagram showing a usage state of a straw harvesting apparatus according to an embodiment of the present invention,
FIG. 2 is a cross-sectional view of a straw harvesting apparatus according to an embodiment of the present invention,
FIG. 3 is a schematic view showing a spore sorting system from a Kim, Yong Sik harvesting apparatus according to the present invention.
FIG. 4 is a flowchart showing a method for selecting spores from a rice paddy field apparatus according to an embodiment of the present invention.

Hereinafter, preferred embodiments of the present invention will be described in detail with reference to the accompanying drawings.

FIG. 1 is a use state view showing a straw harvesting apparatus according to an embodiment of the present invention, and FIG. 2 is a sectional view showing a straw harvesting apparatus according to an embodiment of the present invention.

Referring to FIGS. 1 and 2, a pickling apparatus 100 for pickling a paddy is provided with a water tub provided with a rotary frame 10 wrapping around a Kim Bald 10 wrapped around a net, And hooking rods 2 are installed therebetween. On the hooking rods 2, hanging ropes attached with shells 1 for implanting the spores 50 are hanged.

The water tank filled with brine is made of steel, stainless steel or cement so that the top is opened. And an ordinary rotating frame 10 which is rotated by a predetermined driving device is mounted on one side of the upper surface of the water tub. At this time, the cultured spores 50 are attached to the rotary frame 10, and a Kimbal bridge 20 is installed for installation in a farm. Thereafter, in the lateral space portion of the rotary frame 10, the underwater strings having the shells 1 attached thereto are vertically received in the water tub.

That is, the frame is fixed to the water tub on both sides of the space portion in the direction of the rotary frame 10 side, and a semicircular concave groove is formed on the inner side of both frames at regular intervals. Then, the latching rods 2 are seated on the frames on both sides while being fitted into the respective grooves. Therefore, the water tub is provided with a plurality of hook members (2) horizontally at regular intervals in a space portion other than the rotary frame (10).

The hanging rods 2 are suspended at predetermined intervals. The undergarment is tied at one end to the hook and hangs vertically. A plurality of shells (1) are attached to the underwater line. In order to attach the shell 1 to the underwater line, a hole may be formed in the shell 1 and the under line may be fixed there.

The steerable straw 100 is constructed as follows.

First, the spore 50 extracted from the flesh is transferred to the shell 1 installed in the water tank. At this time, the water tank is filled with a suitable salt water, and a Kimgyegol (20) is wound around the rotary frame (10) installed at the upper part of the water tank and the lower part thereof is suitably immersed in salt water. Then, the driving unit of the rotary frame 10 is driven to rotate the rotary frame 10, and the brine receives physical force to continuously generate a predetermined wave.

As described above, the spore 50 transferred to the shell 1 is cultured for 6 to 7 months at an appropriate water temperature and roughness. The spores 50 are released from the shell 1 by the waves generated by the rotary frame 10 to float in the brine and are implanted into the net wrapped around the rotary frame 10 to perform artificial sieving .

Fig. 3 is a configuration diagram showing a spore sorting system from the Kim Jae-sik harvesting apparatus according to the present invention.

Referring to FIG. 3, the present invention can be configured to include the Kim Jae-sik harvesting apparatus 100, the gimpjob image pickup apparatus 30, and the central management center 200 as described above.

The gimp film imaging apparatus 30 may include a mobile terminal such as a smart phone, and may include a camera unit 32, a communication unit 34, and a control unit 36.

The camera unit 32 preferably includes a special camera capable of recognizing the spore 50 with the naked eye. For example, the camera unit 32 may include a mercury lamp embedded in a biological microscope.

The communication unit 34 may be a communication module for performing wireless communication with the central management center. The communication unit 34 may be a variety of methods including WI-FI, LTE, and the like.

The control unit 36 controls the camera unit and the communication unit as a whole, and receives a signal from the central management unit to adjust the enlargement ratio of the camera and the like.

FIG. 4 is a flowchart showing a method for selecting spores from a rice paddy field apparatus according to an embodiment of the present invention.

Referring to FIG. 4, when the rotary frame is operated in the water tank, the spore attached to the shells is imaged on the net of the Kimgall bridge (S110).

The sprouted spores are photographed by the imaging device (S120), and the photographed images are transmitted to the central control center (S130). The image transmitted to the central control center appears on the terminal screen and measures the amount of spore grafted to the Kimgall bridge. Thereafter, when the amount of the spores measured exceeds the predetermined amount, the selected Kimbal valley is selected (S150). If the measured amount of spores is less than the set amount,), And the spores are again scratched (S140).

On the other hand, the central control center database accumulated due to the processor described above accumulates a large amount of image data in which the seedling planting material is completed, and is converted into big data using the machine learning technique by utilizing the accumulated data, Can be constructed.

The present invention can also be embodied as computer-readable codes on a computer-readable recording medium. A computer-readable recording medium includes all kinds of recording apparatuses in which data that can be read by a computer system is stored. Examples of the computer-readable recording medium include a ROM, a RAM, a CD-ROM, a magnetic tape, a floppy disk, an optical data storage device, and the like, and may be implemented in the form of a carrier wave (for example, transmission via the Internet) .

In addition, the computer-readable recording medium may be distributed over network-connected computer systems so that computer readable codes can be stored and executed in a distributed manner. And functional programs, codes, and code segments for implementing the present invention can be easily inferred by programmers skilled in the art to which the present invention pertains.

While the present invention has been particularly shown and described with reference to exemplary embodiments thereof, it is to be understood that the invention is not limited to the disclosed exemplary embodiments, but, on the contrary, It will be understood by those skilled in the art that various changes and modifications may be made by those skilled in the art without departing from the spirit and scope of the present invention.

1: shell 2:
10: Rotating frame 20:
30: Image pickup device 32:
34: communication unit 36:
100: Kim, Yong-sik Dragon mill 200: Central management center

Patent Citations (3)

Publication number	Priority date	Publication date	Assignee	Title
KR100594729B1	2005-05-06	2006-07-03	이상화	Water harvesting device and seeding method for laver farming
KR101704970B1 *	2016-05-20	2017-02-09	구골홀딩스 주식회사	Eel nursery device
KR20170032153A *	2015-09-14	2017-03-22	(주)빌리언이십일	Control method and apparatus of The ecological environment of the fish farm
Family To Family Citations				

* Cited by examiner, † Cited by third party

Cited By (4)

Publication number	Priority date	Publication date	Assignee	Title
KR20240069259A *	2022-11-11	2024-05-20	대양에스씨 주식회사	Trading platform throughout verification and evaluation of laver seed
Family To Family Citations				
KR20230030147A	2021-08-25	2023-03-06	최옥산	Device for picking laver farming
KR102478888B1	2022-06-13	2022-12-16	최옥산	Device for picking laver farming
KR102478878B1	2022-09-23	2022-12-16	최옥산	Device for picking laver farming

* Cited by examiner, † Cited by third party, ‡ Family to family citation

Similar Documents

Publication	Publication Date	Title
KR20190055413A	2019-05-23	Methods for selecting spores from laver farm devices
KR101598898B1	2016-03-03	The system for precaution for paralichthys olivaceus disease using and an analysis of image and sound
JP6842100B1	2021-03-17	Aquatic animal detection device, information processing device, terminal device, aquatic animal detection system, aquatic animal detection method, and aquatic animal detection program
JP2019170349A	2019-10-10	Aquatic production management system
KR101540892B1	2015-07-30	Height Adjustable Seed Collector for Scapharca subcrenata
CN116681243B	2023-12-01	Intelligent agricultural equipment management system and method based on Internet of things
JP2019104420A	2019-06-27	Offshore wind power generation facility
CN115526521A	2022-12-27	Plant growth state monitoring and alarming system for plant factory
CN116152718A	2023-05-23	An intelligent observation device and observation method for shrimp farming
CN210352708U	2020-04-21	Coastal zone breeding protection area monitoring system
CN107242176A	2017-10-13	A kind of cage culture system
CN106483128A	2017-03-08	Plant culture supports that method is supported in device and plant culture
CN113567647A	2021-10-29	A rapid monitoring system and control method for submerged plant community
KR20160044817A	2016-04-26	Greenhouse control system and control method thereof
JP4820966B2	2011-11-24	Liquid supply control device
JPWO2018216212A1	2019-06-27	Network management system, method and program
CN115379150B	2023-03-14	System and method for automatically generating dynamic video of rice growth process in remote way
US11877549B2	2024-01-23	Controller for seaweed farm
Sabri et al.	2018	Precision crop management for indoor farming
KR102368811B1	2022-02-28	Smart seaweed cultivation management system and management method
CN104969885B	2017-11-17	A kind of marine cage fish culture system and method
KR100594729B1	2006-07-03	Water harvesting device and seeding method for laver farming
KR101933630B1	2018-12-28	Smart Orchard System for Comprehensive Management of Bare Ground Orchard
CN116840166A	2023-10-03	Method for monitoring drought conditions of drip irrigation under cotton film based on multispectral remote sensing detection
KR102824870B1	2025-06-24	Method for transplanting zostera using the zostera restoration system

Priority And Related Applications

Priority Applications (1)

Application	Priority date	Filing date	Title

KR1020170152108A	2017-11-15	2017-11-15	Methods for selecting spores from laver farm devices

Applications Claiming Priority (1)

Application	Filing date	Title
KR1020170152108A	2017-11-15	Methods for selecting spores from laver farm devices

Legal Events

Date	Code	Title	Description
2017-11-15	A201	Request for examination	
2017-11-15	PA0109	Patent application	**St.27 status event code**: A-0-1-A10-A12-nap-PA0109
2017-11-15	PA0201	Request for examination	**St.27 status event code**: A-1-2-D10-D11-exm-PA0201
2017-11-24	P11-X000	Amendment of application requested	**St.27 status event code**: A-2-2-P10-P11-nap-X000
2017-11-24	P13-X000	Application amended	**St.27 status event code**: A-2-2-P10-P13-nap-X000
2018-10-12	D13-X000	Search requested	**St.27 status event code**: A-1-2-D10-D13-srh-X000
2018-12-07	D14-X000	Search report completed	**St.27 status event code**: A-1-2-D10-D14-srh-X000
2019-03-19	E902	Notification of reason for refusal	
2019-03-19	PE0902	Notice of grounds for rejection	**St.27 status event code**: A-1-2-D10-D21-exm-PE0902
2019-05-20	T11-X000	Administrative time limit extension requested	**St.27 status event code**: U-3-3-T10-T11-oth-X000
2019-05-23	PG1501	Laying open of application	**St.27 status event code**: A-1-1-Q10-Q12-nap-PG1501
2019-06-19	E13-X000	Pre-grant limitation requested	**St.27 status event code**: A-2-3-E10-E13-lim-X000
2019-06-19	P11-X000	Amendment of application requested	**St.27 status event code**: A-2-2-P10-P11-nap-X000
2019-06-19	P13-X000	Application amended	**St.27 status event code**: A-2-2-P10-P13-nap-X000
2019-10-08	E701	Decision to grant or registration of patent right	
2019-10-08	PE0701	Decision of registration	**St.27 status event code**: A-1-2-D10-D22-exm-PE0701
2019-10-14	GRNT	Written decision to grant	
2019-10-14	PR0701	Registration of establishment	**St.27 status event code**: A-2-4-F10-F11-exm-PR0701
2019-10-14	PR1002	Payment of registration fee	**St.27 status event code**: A-2-2-U10-U11-oth-PR1002 **Fee payment year number**: 1
2019-10-18	PG1601	Publication of registration	**St.27 status event code**: A-4-4-Q10-Q13-nap-PG1601
2022-07-18	PR1001	Payment of annual fee	**St.27 status event code**: A-4-4-U10-U11-oth-PR1001 **Fee payment year number**: 4
2023-09-02	PR1001	Payment of annual fee	**St.27 status event code**: A-4-4-U10-U11-oth-PR1001 **Fee payment year number**: 5
2023-10-13	R18-X000	Changes to party contact information recorded	**St.27 status event code**: A-5-5-R10-R18-oth-X000
2024-08-12	PR1001	Payment of annual fee	**St.27 status event code**: A-4-4-U10-U11-oth-PR1001

			Fee payment year number: 6
2025-08-28	PR1001	Payment of annual fee	**St.27 status event code**: A-4-4-U10-U11-oth-PR1001 **Fee payment year number**: 7
2025-08-28	U11	Full renewal or maintenance fee paid	**Free format text**: ST27 STATUS EVENT CODE: A-4-4-U10-U11-OTH-PR1001 (AS PROVIDED BY THE NATIONAL OFFICE) **Year of fee payment**: 7

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